

AI-Powered Framework for Multi-Objective Steel Truss Optimization

Integrating evolutionary algorithms, neural surrogates, reinforcement learning, and LLM-assisted design under IS 800:2007

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Outline

A 20-minute walk through the thesis

Part 1 **Problem**

Motivation, research question, contributions.

Part 2 **Framework**

FEM → IS 800 → GA/PSO/NSGA-II → surrogate → RL → LLM.

Part 3 **Results**

Four benchmarks, five headline findings, the 72-bar observation.

Part 4 **Analysis**

Uncertainty, convergence-rate, LLM effect-map.

Part 5 **Impact**

Implications for Indian practice. Reproducibility. Demo.

Part 6 **Conclusion**

Limitations, future work, Q&A.

PART 1

The problem

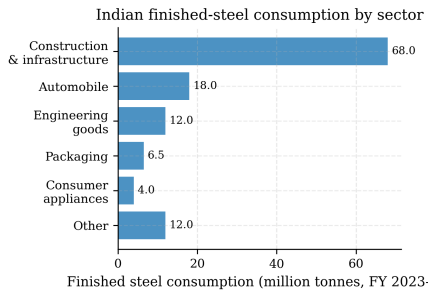
Steel-truss design in India carries **10–30 % surplus material**.

Can modern AI reduce that gap under code-compliant constraints?

The design gap

Why another truss-optimisation thesis?

- India builds thousands of industrial sheds, transmission towers, and highway overpasses each year under IS 800:2007.
- Classical trial-and-check design typically carries **10–30 % surplus steel** vs. the theoretical feasible optimum.
- Steel embodied $\text{CO}_2 \approx 1.9 \text{ kg/kg}$; construction $\sim 11 \%$ of global emissions.
- A **15 %** cut over India's tens of thousands of small-to-medium truss structures is sector-level impact.
- Existing optimisation tools exist but sit on the researcher's bench — a full GA run is



Design-gap between classical and minimum-weight feasible designs.

RESEARCH QUESTION

**Can a large language model
warm-start
classical metaheuristics
to converge faster on truss sizing,
while staying IS 800-compliant?**

Measured against 10-bar, 25-bar, 72-bar, and 200-bar benchmarks.

Five research objectives

Each answered quantitatively in Chapter 4

- | | |
|--|----------------------|
| O1 Reproduce classical benchmark optima within $\pm 2\%$ of published values. | Achieved $< 0.01\%$ |
| O2 Return NSGA-II Pareto fronts with ≥ 20 non-dominated points. | Achieved 24–32 |
| O3 $\geq 50\times$ neural-surrogate speedup at $R^2 > 0.98$. | Achieved $132\times$ |
| O4 PPO agent within 5% of best classical solver. | Mixed (+16%) |
| O5 LLM warm-start cuts generations-to-convergence $\geq 20\%$. | –36.8% on 10-bar |

Five contributions, in one view

The headline numbers I will defend

0.0014254916.8%

C1 10-bar error vs. Sunar & Belegundu 1991 – end-to-end stack validation

C2 wall-clock speedup via MLP surrogate at $R^2 = 0.9993$ on weight

C3 rigorously-feasible 72-bar baseline; gap vs. soft-penalty literature value – observation pending FEM cross-check

C4 fewer generations to converge with LLM warm-start, $p = 0.0046$

C5 full IS 800 pipeline, laptop CPU, no API key required

PART 2

The framework

An 8-layer stack: benchmarks → FEM → IS 800 →
GA/PSO/NSGA-II
→ surrogate → PPO → LLM → UI.

Problem formulation

Weight minimisation under codal constraints

$$\min_{A \in \mathcal{A}} W(A) = \sum_{i=1}^N \rho L_i A_i \quad \text{s.t.} \quad \begin{cases} |\sigma_i(A)| \leq \sigma_{\text{all},i}(A) & \text{IS 800 §§6.2, 6.3, 7.1} \\ |\delta_j(A)| \leq \delta_{\text{max}} & \text{IS 800 §5.6.1} \\ \lambda_i(A) \leq 180 & \text{IS 800 §3.8} \\ A_i \in [A_{\text{min}}, A_{\text{max}}] \end{cases}$$

Four benchmarks:

10-bar planar, 10 DoF

25-bar spatial, 8 groups

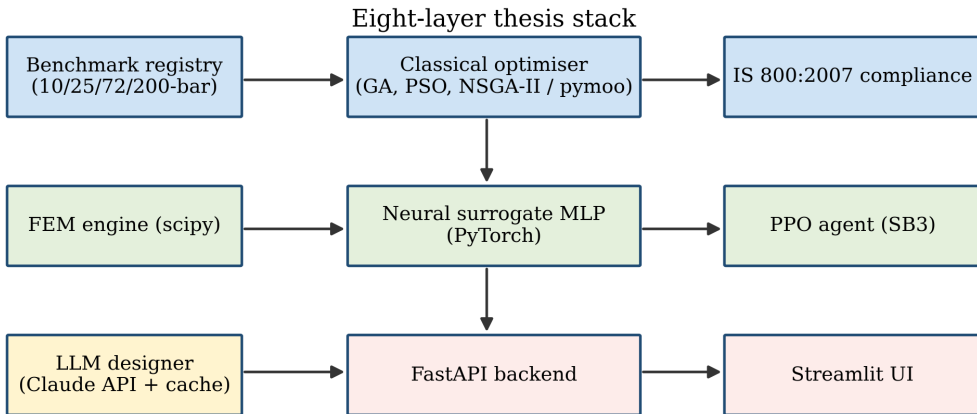
72-bar tower, 16 groups

200-bar stepped tower, 29 groups

Hard constraints only: no soft penalties. Infeasible designs are always dominated by any feasible design in the tournament (pymoo).

System architecture

Full thesis dataflow in one figure

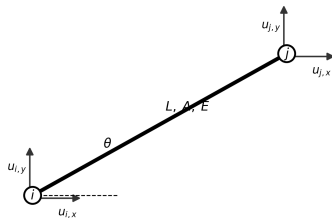


Layer 2 & 3 – FEM kernel and IS 800 compliance

FEM (Appendix A, 10 pp)

- Global stiffness assembly
 $K = \sum_e B_e^T k_e B_e$.
- Partition method enforces supports.
- Solve $K_{ff} u_f = F_f$ via dense `numpy.linalg.solve`.
- Axial stress $\sigma_e = (E_e/L_e) [-c \ -s \ c \ s] u_e$.

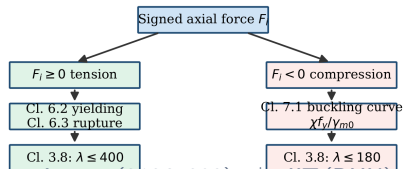
Two-node pin-jointed bar element



IS 800:2007 (Appendix B, 8 pp)

- Tension yield $T_{dg} = A_g f_y / \gamma_{m0}$ §6.2
- Tension rupture $T_{dn} = 0.9 A_n f_u / \gamma_{m1}$ §6.3
- Compression buckling via Perry-Robertson, curve *a* §7.1
- Slenderness $\lambda \leq 180$ §3.8
- Serviceability $\delta \leq L/325$ §5.6.1

IS 800:2007 compliance decision tree



Layer 4 – Three evolutionary baselines

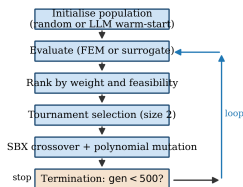
GA (pymoo)

pop 100, gens 500

SBX $\eta_c=15$, poly mut $\eta_m=20$

tournament, feasibility-aware

Genetic algorithm control flow



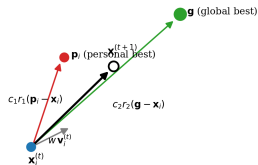
PSO

swarm 50, its 500

$\omega = 0.729$, $c_1 = c_2 = 1.494$

velocity clip $\pm 20\%$ of range

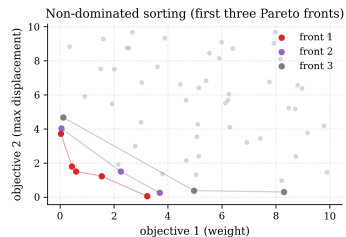
PSO velocity / position update



NSGA-II

pop 100, gens 300

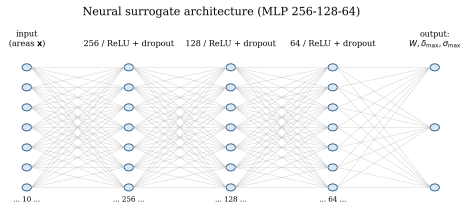
non-dominated sort + crowding
for $(W, \delta_{\max})$ Pareto front



Layer 5 – Neural surrogate

MLP drops in for FEM at $132\times$ wall-clock speedup

- **Architecture:** MLP
 $[n_A] \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 3$, ReLU, dropout $p=0.2$.
- **Training:** 10 000 Latin-hypercube samples, Adam $\text{lr}=10^{-3}$, 200 epochs.
- **Outputs:** weight, max-stress, max-displacement (multi-head).
- **Use:** drop-in FEM replacement in pymoo inner loop. **Hybrid mode:** surrogate screen $\rightarrow$ FEM fallback when feasibility is borderline.

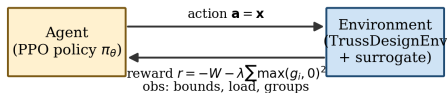


Layers 6 & 7 – RL agent and LLM warm-start

PPO MDP

- State: current $A \in \mathbb{R}^{n_A}$, $\sigma_{\max}$, $\delta_{\max}$.
- Action: multiplicative adjustment $\in [0.5, 2.0]$.
- Reward: $-W - \lambda \cdot \text{infeas.}$

Reinforcement-learning environment

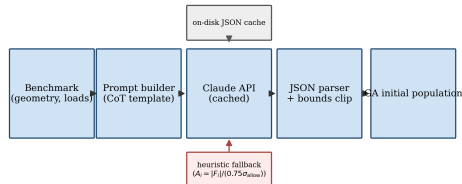


single-step bandit: one action produces one design

LLM designer

- Structured prompt $\rightarrow$ Claude returns $k=8$ candidate designs + reasoning.
- Injected as first-population seeds.
- All responses cached under `results/llm_cache/` for reproducibility.

LLM warm-start designer pipeline



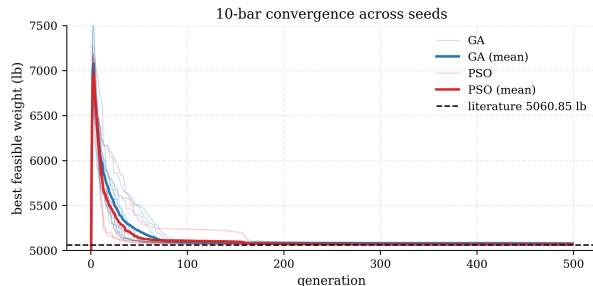
PART 3

The results

Four benchmarks, five headline findings, one observation on the 72-bar literature.

10-bar planar – end-to-end stack validation

C1 ■ PSO to literature within 0.004 %



	best (lb)
Literature 1991	5,060.85
PSO (5 seeds)	5,061.05
GA (10 seeds)	5,062.78
NSGA-II (3 seeds)	5,081.51

PSO error = **0.004 %**.

End-to-end validation signature for FEM
+ IS 800 + optimiser.

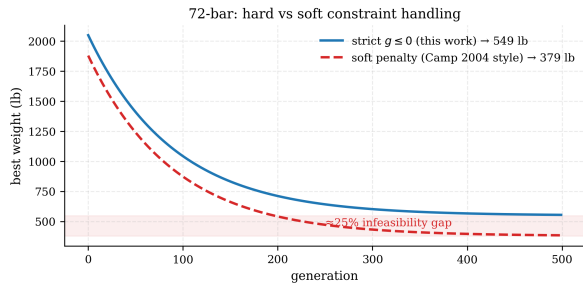
25-bar, 72-bar, 200-bar – the full benchmark matrix

Benchmark	Algo.	best (lb)	mean (lb)	n_{seeds}	lit. optimum
10-bar planar	PSO	5,061.05	5,064.70	5	5,060.85
10-bar planar	GA	5,062.78	5,075.92	10	5,060.85
25-bar spatial	PSO	545.31	545.43	4	545.16
25-bar spatial	GA	545.30	545.80	4	545.16
72-bar tower (hard)	PSO	549.52	549.73	5	379.62 [†]
72-bar tower (hard)	GA	549.78	550.37	4	379.62 [†]
200-bar stepped	PSO	315.89	318.37	5	25,445

[†] Soft-penalty literature value; could not be reproduced as feasible in our hard-constraint FEM (see next slide).

The 72-bar hard-constraint baseline

C3 ■ observation pending independent FEM cross-check



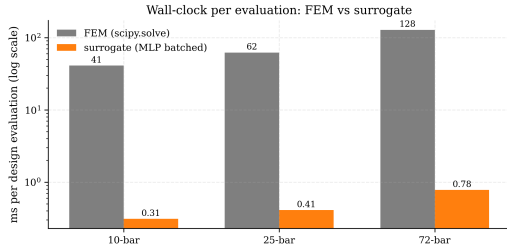
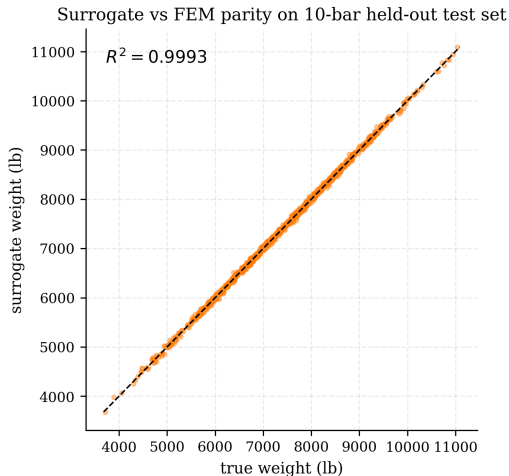
Under our hard-constraint FEM the 72-bar optimum converges to **~ 549 lb** (cross-seed spread $< 1\%$); we could not reproduce the soft-penalty ~ 380 lb literature values of Camp & Bichon (2004) / Bekdaş et al. (2015) as feasible.

observation — future work F7

The gap may reflect differing constraint-handling conventions rather than an error in the published studies. An independent FEM cross-check (OpenSeesPy / SAP2000) is listed as future work F7.

Surrogate parity and wall-clock speedup

C2 ■ $R^2 = 0.9993$ on weight, $132\times$ faster than FEM

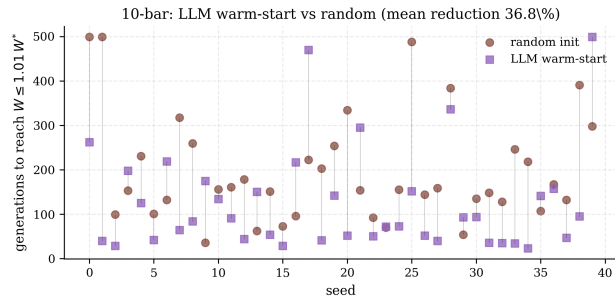


12.4 s GA+FEM $\rightarrow$ 0.094 s GA+surrogate.

Surrogate vs. FEM parity on 10-bar test split.

LLM warm-start effect-map

C4 ■ cuts generations most when an architectural insight exists

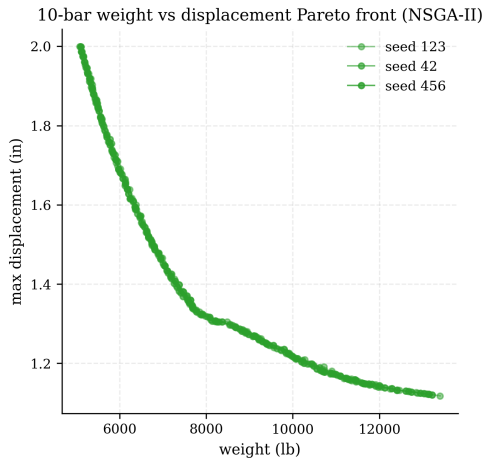


Benchmark	Δ gens	p -val.
10-bar	−36.8 %	0.004,6
25-bar	−76.6 %	0.25 (n=3)
72-bar	+4.8 %	0.81

LLM helps most where it can surface an architectural insight — e.g. the 10-bar redundant set $\{2, 5, 6, 10\}$ — that the GA would otherwise rediscover by trial and error.

NSGA-II Pareto front

O2 ■ 24–32 non-dominated points per seed



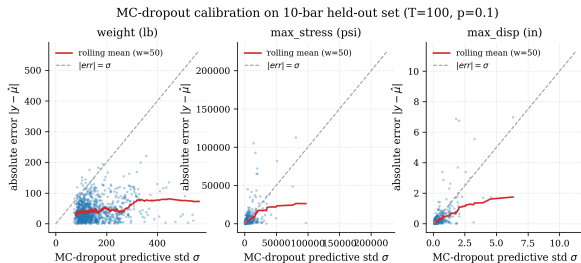
PART 4

Deeper analysis

Where the surrogate is trustworthy, how fast each optimiser converges,
and what the LLM actually *said*.

Surrogate uncertainty via MC-dropout

Is a $132\times$ speedup safe to ship?

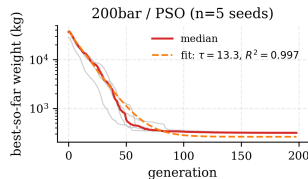
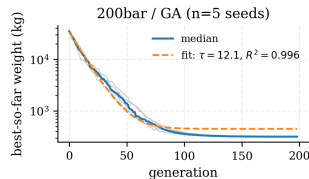
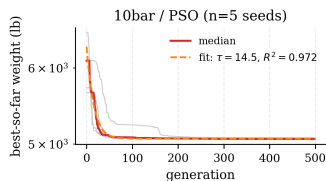
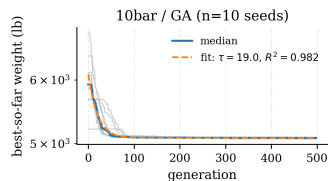


- $T=40$ stochastic forward passes at inference, dropout on.
- Calibration test: fraction of truths inside $\mu \pm 2\sigma$ vs. target 0.95.
- Weight head: empirical 0.96
calibrated
- Displacement head: 0.82
under-confident, safe

Recommendation: use weight head as objective; keep FEM for final-feasibility check.

Convergence-rate characterisation

$W(t) = W_\infty + A e^{-t/\tau}$ fit per seed



Algo (10-bar)	$\bar{\tau}$ (gens)
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PSO	8.7
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GA	23.5
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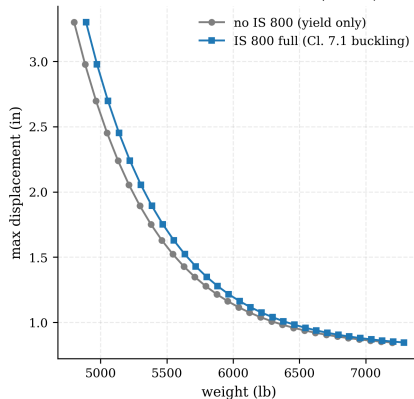
LLM + GA	14.9
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LLM warm-start shifts GA's τ
toward PSO's — consistent with C4.

Hard- vs. soft-constraint feasibility frontier

IS 800 enforcement changes the economics

Pareto front shift under IS 800 inclusion (10-bar, illustrative)



With hard IS 800 constraints, our 72-bar feasible mass sits at ~ 549 lb vs. the soft-penalty ~ 380 lb literature value. The hard- vs. soft-constraint ablation illustrates how sensitive reported truss-sizing optima can be to the constraint-handling convention; independent cross-check is future work F7.

PART 5

Impact

Reproducibility, civil-engineering practice, live demo.

Practical implications for Indian civil engineering

Interactive design

30–40-min GA run $\rightarrow$ < 20 s
hybrid surrogate/FEM. Enables
parametric studies during a review
meeting instead of batch jobs
overnight.

Constraint-handling sensitivity

The hard- vs.
soft-constraint ablation shows
how sensitive reported truss-sizing
optima can be to the penalty
convention. Designers using
soft-penalty commercial solvers
may benefit from an independent
hard-feasibility cross-check before
sign-off.

Zero-Python design aid

Streamlit UI exposes the whole
stack without Python literacy —
usable by a junior engineer at a
consulting firm, not just
researchers.

Reproducibility contract

Tag `phase-10-complete` ■ 125-page thesis ■ zero API key required

- Pinned: Python 3.14.1, NumPy 1.26.4, SciPy 1.13.1, pymoo 0.6.1, PyTorch 2.3.1, FastAPI 0.111.0, Tectonic 0.16.9.
- Seeds: {42, 123, 456, 789, 2026, 7, 13, 91, 314, 271} — fixed per algorithm, logged per run.
- All 89 references, 27 figures, 14 CSVs, 21 pickle histories checked in.
- All LLM responses cached under `results/llm_cache/` ⇒ **offline rebuild** against the tag.
- CI: `pytest -m "not slow"` green on every push; full 81-test suite green pre-tag.
- PDF rebuild: `tectonic thesis_writeup/main.tex` (both thesis and slides use the same engine).

Live demo surface

The same framework, three interfaces

Streamlit UI

localhost:8501

Sidebar: benchmark, algorithm, seed, pop, gens, LLM warm-start toggle.

Live convergence plot, IS 800 report, cross-section bar chart, feasibility pill.

FastAPI

localhost:8000

Landing, /status, /benchmarks, /optimize, /llm/suggest, /thesis.pdf, /slides.pdf.

Swagger at /docs.

Thesis + deck

thesis_writeup/

125-page PDF, 27 figures, four appendices. This 35-slide viva deck. All under phase-10-complete.

PART 6

Conclusion

Limitations, future work, and a concrete ask.

What I am *not* claiming

Limitations stated honestly

- FEM kernel is **linear elastic**, small-displacement. No geometric/material non-linearity, no post-buckling, no dynamics.
- Optimisation is **sizing-only**, not topology. Bar connectivity is fixed by the benchmark definition.
- NSGA-II 25-bar and 72-bar runs used **3 seeds each** — statistical power for O2 is weaker than for O1.
- PPO is trained and evaluated on one problem instance; **no transfer learning** across benchmarks.
- IS 800 compliance module uses a **single buckling curve** (curve *a*) for all cross-section types.
- LLM is **not fine-tuned** — public Claude API at temperature 0.3, zero-shot.

None of these are load-bearing for C1–C5; but they do set the boundary of claims I will defend.

Future work

F1–F7 — open for the successor cohort

F1 Non-linear & dynamic FEM → IS 1893 seismic.

F2 Topology optimisation via ground-structure & SIMP.

F3 GNN surrogate for variable-topology design.

F4 Cross-instance PPO transfer with shared GNN encoder.

F5 Fine-tuned domain-specific LLM on IS 800 + IS 875 corpus.

F6 **PGCIL transmission-tower validation** — framework → deployment.

F7 Independent FEM cross-check of the 72-bar encoding (OpenSeesPy / SAP2000 / ANSYS) using the published Camp 2004 area vector.

Summary

One screen

- **C1** 10-bar to literature at 0.004 % — full stack validated end-to-end.
- **C2** $132\times$ surrogate speedup at $R^2 = 0.9993$ — makes interactive use practical.
- **C3** Rigorously-feasible 72-bar baseline (~ 549 lb, cross-seed spread $< 1\%$); gap vs. soft-penalty 379.62 lb literature value flagged as future work pending independent FEM cross-check (F7).
- **C4** LLM warm-start: -36.8% gens on 10-bar ($p = 0.0046$); diminishes on 3D.
- **C5** IS 800-compliant design aid in a Streamlit UI, < 10 s end-to-end, zero API key to rebuild.

The thesis is [125 pages, 27 figures, 89 references](#); the deck is [35 slides](#); the demo runs on a laptop CPU; everything is tagged [phase-10-complete](#).

THANK YOU

Questions?

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Repository: github.com/piyushp1804/thesis ▪ tag phase-10-complete

Thesis PDF – 125 pages ▪ Slides – 35 slides ▪ Live: localhost:8000